Ex.no: 14 Date:

Packet sniffing using raw sockets

Aim:

To implement packet sniffing using raw sockets in python, to capture and analyze network packets

Program:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
 if IP in packet:
    IP_layer = packet[IP]
    protocol = IP_layer.proto
    src_ip = ip_layer.src
    dst_ip = ip_layer.dst
    protocol_name = ""
    if protocol == 1:
        protocol_name = "ICMP"
    elif protocol == 6:
        protocol_name = "TCP"
    elif protocol == 17:
        protocol_name = "UDP"
    else:
        protocol_name = "Unknown protocol"
```

Protocol : UDP

source IP : 192.168.1.1

destination. IP : 192.168.1.6

Result :

Thus, packet sniffing using raw sockets was implemented successfully and network packets were captured and analyzed.